

Mapping The Warp of the Milky Way Galaxy

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Abstract

Utilizing 3 hours of observation time, my lab partners and I collected data on the power spectrum of the High Velocity Clouds in the Milky Way galaxy using the radio telescope at the Leuschner Observatory. Our pointing ranged from $l = 75^\circ$, $b = +20^\circ \rightarrow -20^\circ$ with each pointing collected with an integration time of about 4 minutes. After converting frequency information into velocities and distances, we convolved a Gaussian profile as the kernel to smooth the data and also removed any noise below temperatures of 0K. Using the Python module 'PyLab', we created an image plotting the distance from and height above the galactic center, mapping temperatures to color intensities. The result of the experiment shows an uneven distribution of H1 emissions above and below the galactic plane.

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1 Introduction

1.1 The Leuschner Radio Telescope

The radio telescope used for in this lab is located at the Leuschner Observatory at a latitude of $+37.8732^\circ$ and longitude of $-122^\circ 9'$ in Berkeley, California. The Leuschner Observatory is a student observatory ran by the Astronomy Department of the University of California at Berkeley. The diameter of the dish is approximately 4.5m and operates at a frequency range from 1320–1740MHz with a 110MHz bandwidth. The telescope operates with mixed local oscillator signals transferred to an ADC after passing through two bandpass filters in order to minimize contaminations from other signal frequencies.

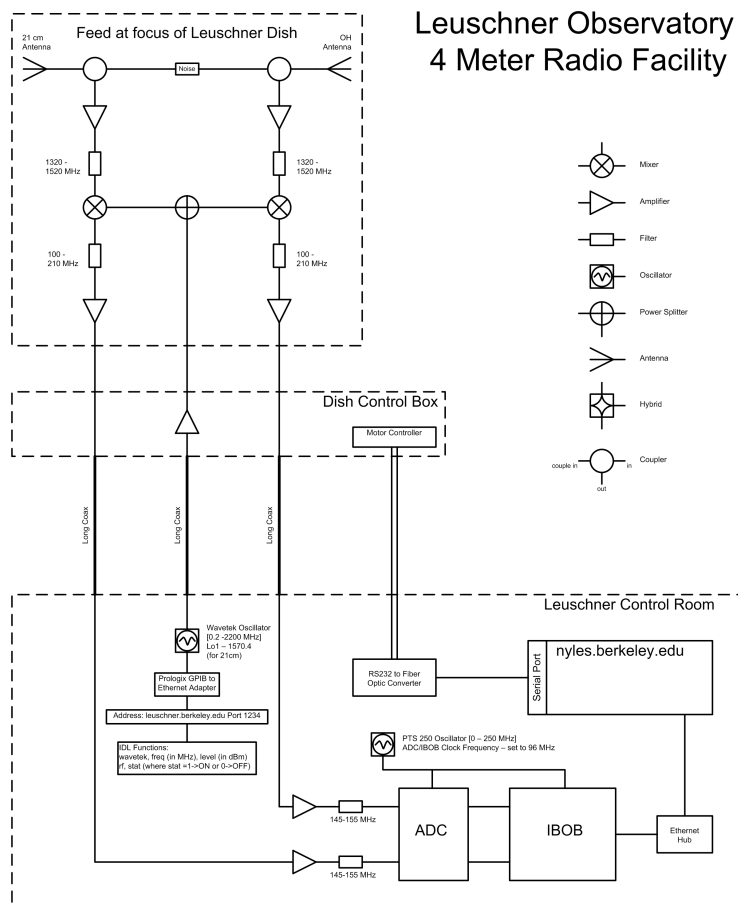


Figure 1: This is a picture of the circuit diagram schematics for the 4.5m diameter radio telescope at the Leuschner Observatory in Berkeley, California. (Image was taken from http://ugastro.berkeley.edu/radio/leuschner/Leuschner_circuit.png on May 4, 2014.)



Figure 2: This is a picture of the 4.5m diameter radio telescope at the Leuschner Observatory in Berkeley, California ($-122^{\circ}9'$, $+37.8732^{\circ}$). Wiring runs from the telescope to the near by building underground where they are processed through the electronics required for operation. (Image was taken from <http://ugastro.berkeley.edu/pictures/science/leuschner/radio/dish.jpg> on May 4, 2014.)

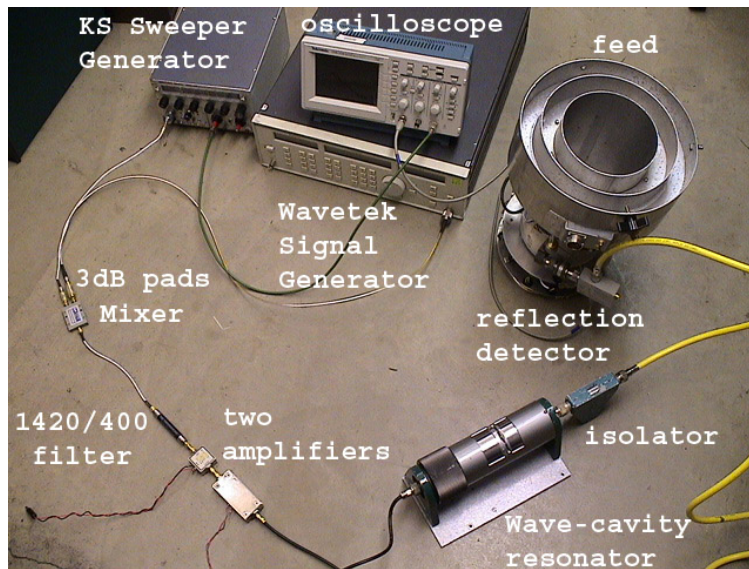


Figure 3: This is a picture of the electronics used to operate the 4.5m radio telescope at the Leuschner Observatory along with the actual reflector detector of the telescope itself. (Image was taken from http://ugastro.berkeley.edu/pictures/science/leuschner/radio/feedmatch_equipment_writing.jpg on May 4, 2014.)

1.2 Mapping The Galactic Warp

External galaxies are often distorted by nearby galactic neighbors. To determine if the Milky Way galaxy is distorted or 'warped', my partners and I used the radio telescope at the Leuschner Observatory and profiled a section of the sky vertically slicing across the galactic plane. Hypothetically, a 'flat' galaxy would have emission peaks everywhere at galactic latitude $b = 0^\circ$. To observe and display these distortions, we were only able to take one complete vertical profile of the Milky Way. If our galaxy does experience a warp, the result of this experiment should show an uneven distribution of H1-emissions along galactic latitudes above and below $b = 0^\circ$.

2 Background Theory

2.1 Hydrogen 21cm Line

The intrinsic spin property of the electron in a Hydrogen causes the atom to emit a photon at precisely a distance known as the 21cm line or a frequency of 1420.4MHz. Additionally, the transition between the three-fold degenerate triplet states and the non-degenerate singlet state in Hydrogen atoms have an energy difference $\Delta E = h\nu \approx 5.874 \times 10^{-9} eV$. This energy difference is transmitted by photons when the spin state of the electron flip from their alignment with the spin state of the proton to an anti-aligned position. Since the frequency of 1420.4MHz is in the radio wave band of the light spectrum, H1-emissions penetrate most of the dust along the line-of-sight and are not absorbed or scattered. Utilizing the 21cm line, radio astronomers can examine spectra line emissions and study the propagation of H1-emitted photons across the vacuum of space.

2.2 The Doppler Effect

The Doppler effect is the change in frequency of a wave or periodic event for an observer in motion relative to a source. An everyday example of the doppler effect can be examined when a vehicle emitting a siren or horn approaches, passes, and then recedes away from an observer. As the sound waves approach, the frequency of the sound increases, making the pitch of the sound also higher. As the vehicle recedes, the frequency and pitch diminishes. This is because each wave takes slightly less time to reach the observer than the previous. For light waves, the effect is the same. While photons travel through space, the color or frequency of the wave also stretches and contracts, making the waves "bunch together". On the contrary, if the wave source is moving away from the observer, the waves "spread out". This "bunching" and "spreading" of propagating light waves is known as red and blue shift.

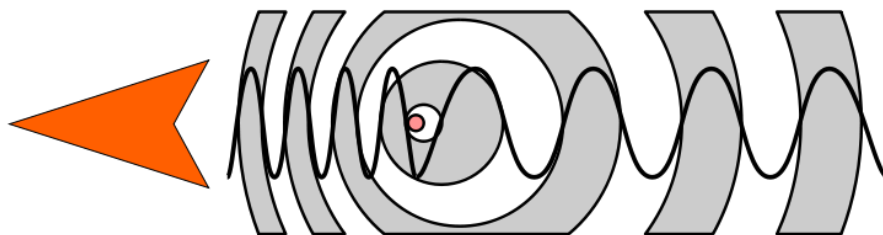


Figure 4: As waves propagate across space, the wavelength resulting from the motion of the source will either stretch or contract depending on its direction of motion. (Image was taken from http://en.wikipedia.org/wiki/File:Doppler_effect_diagrammatic.svg on May 4, 2014.)

Additionally, the velocity of the observer and the source are determined by the medium through which the waves are transmitted. The total Doppler effect therefore results from the motion of the source, the observer, and the medium and each of these effects must be analyzed separately. To find the Doppler velocity of light waves, we can use Equation 1 shown below. In this equation, $\Delta\nu = \nu_{obs} - \nu_0$ and c is the speed of light. However, for waves which do not encounter much of a medium like light from point sources, only the relative difference in velocities between the observer and the source have to be considered.

$$v_{doppler} = - \left(\frac{\Delta\nu}{\nu_0} \right) c \quad (1)$$

2.3 Galactic Coordinate Transformation

The galactic coordinate system is an astronomical coordinate system based on the position of the sun about the center of the Milky Way. Galactic longitude (l) measures the angular distance of an object eastward along the galactic equator from the galactic center. Galactic latitude (b) measures the angular distance of an object above the galactic equator with positive to the north and negative to the south. These coordinates are analogous to terrestrial longitude and latitude and are usually measured in degrees.

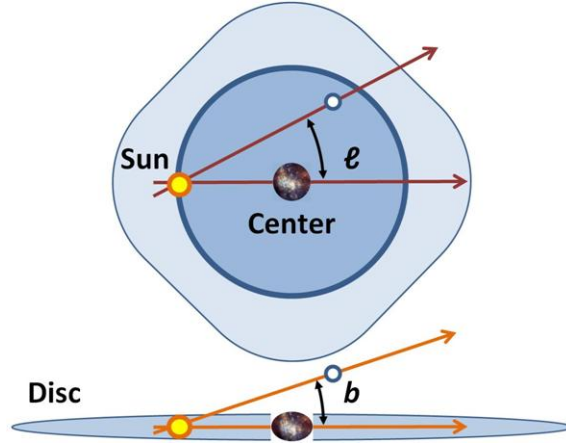


Figure 5: In the galactic coordinate system, because of the large distances, the Sun is placed at the origin. Galactic longitude (l) is measured from the direction from the Sun to the center of the galaxy on the galactic plane. Galactic latitude (b) measures the angle above the galactic plane. Additionally the region between the sun's orbit and the galactic center is known as the solar circle. (Image was taken from http://en.wikipedia.org/wiki/File:Galactic_coordinates.JPG on May 5, 2014.)

In spherical coordinates, the position of stars are often expressed in right ascension and declination without use of a coordinate for distance. Because celestial objects are extremely far away, astronomers often have little to no information about their exact distances and therefore use only the object's direction. Additionally, the direction of fairly distant objects is the same for all observers and it is convenient to specify this direction with the same coordinates for every object. Using Equation 2 shown below, we can convert from terrestrial latitude and longitude to spherical coordinates (α, δ).

$$\vec{x} = \begin{bmatrix} \cos(lat) \times \cos(long) \\ \cos(lat) \times \sin(long) \\ \sin(lat) \end{bmatrix} \quad (2)$$

The horizontal coordinate system is a celestial coordinate system that utilizes an observer’s local horizon as it’s fundamental plane. This coordinate system divides the sky into an upper hemisphere and a lower hemisphere. In the upper hemisphere, objects and stars are visible or within sight while in the lower hemisphere, the our view of the sky is blocked by the Earth. The “great circle” separating hemispheres is called celestial horizon or rational horizon and the pole of the upper hemisphere is called the zenith. Using the spherical coordinates (α, δ) and multiplying it by the rotational matrix shown in Equation 3, coordinates in terrestrial latitude and longitude can be converted into celestial coordinates (az, alt) . Additionally, this depends on the observer’s terrestrial latitude ϕ .

$$R_{(\alpha, \delta) \rightarrow (az, alt)} = \begin{bmatrix} -\cos(LST) \times \sin(\phi) & -\sin(LST) & \cos(LST) \times \cos(\phi) \\ \sin(LST) \times \sin(\phi) & -\cos(LST) & -\sin(LST) \times \cos(\phi) \\ \cos(\phi) & 0 & \sin(\phi) \end{bmatrix} \quad (3)$$

Because the Milky Way galaxy is extremely large and takes a long time to complete a full rotation, galactic coordinates do not change much on “Earth-based” scales of time. To convert terrestrial coordinates to galactic, we can utilize the rotational matrix shown in Equation 4. By multiplying spherical right ascension and declination coordinates by the rotational matrix shown below in Equation 4, we can convert (α, δ) coordinates to (l, b) coordinates. Since hour angle and declination are “Earth-based” coordinates, this conversion will only depend on the observer’s terrestrial latitude.

$$R_{(\alpha, \delta) \rightarrow (l, b)} = \begin{bmatrix} -0.054876 & -0.873437 & -0.483835 \\ 0.494109 & -0.444830 & 0.746982 \\ -0.867666 & -0.198076 & 0.455984 \end{bmatrix} \quad (4)$$

3 Methods and Observations

3.1 Operating The Telescope

To operate and control the 4.5m radio telescope at the Leuschner Observatory, my partners and I programmed and remotely ran a Python script on Leuschner’s local computer via SSH. Since the H1-emission line is centered at about 1420.4MHz and is present within a 12MHz Nyquist zone, we chose to set our L.O. frequency to a range 12 Nyquist zones, or 144MHz, away. We also added an additional 6MHz, half the nyquist frequency, to center the H1-line. This made the center emission at a frequency of about 1270.4MHz. For our observations, we also programmed the telescope to take four separate measurements per pointing for the purpose of calibrating our data. The four measurements taken include a sample with the L.O.’s frequency set to 2MHz above the center frequency at 1272.4MHz, $10dB_m$, another with the frequency shifted 2MHz below at 1268.4MHz, $10dB_m$, and two measurements using a noise diode to add $T_{noise} = 100K$ to the total system temperature T_{sys} of each signal. When operating the telescope, we also had to “home” the telescope before each measurement to make sure that it was functioning properly. Homing the dish resets the positioning drives of the telescopes and returns the telescope to its default homing position (alt=90°, az=180°).

3.2 Mapping Area

The mapping area we covered for our observations was limited by the amount of time give to our group to operate the telescope and the pointing limits of the dish itself (see Figure 6).

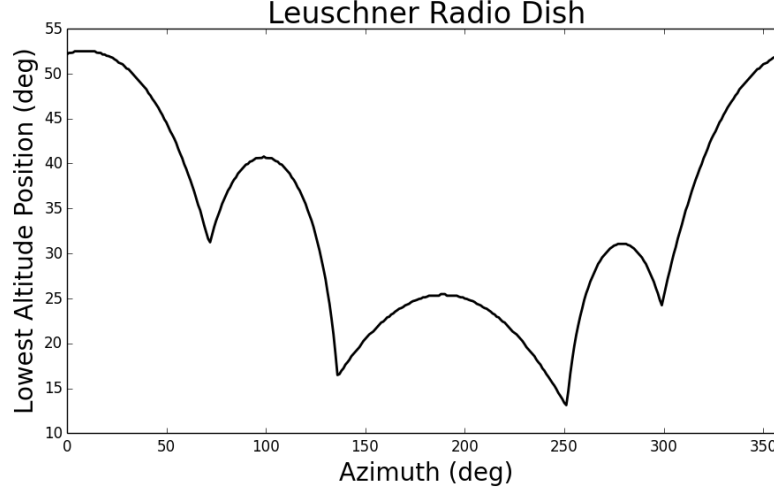


Figure 6: This graph shows the pointing limits of the 4.5m radio telescope at the Leuschner Observatory. Pointing limits result from the boundaries of the telescope and it's location.

After calibrating data taken from May 2, 2014 and May 3, 2014, we realizing that our coordinates were being processed incorrectly and that the telescope did not always take valid measurements per pointing. In the end, the total time range of our accurate observations was diminished to May 4, 2014 from 17:48:37 - 20:41:31 UT. To compensate for this, we set our galactic coordinates to $l = 75^\circ$ and ranged b from $+20^\circ$ to -20° , taking pointings at a spacing of 2° . To ensure that we had all the data from each pointing, we first set our coordinates to cover 4° increments within the range of b . After, we set them to cover 2° increments and then another set of 4° increments, but skewed by 2° . This ensured that we had at least one pointing for every position in our l and b range. Additionally, we set the integration time of each measurement to about 4 minutes per pointing or 8192 data points per sample to ensure that our observation fell within our 3 hour observation window.

3.3 Signal Processing in Python

3.3.1 Manipulating Collected Data

Observational data files collected from the radio telescope at The Leuschner Observatory are stored in a '.log' format. To open these files, I used the "ReadSpec" module provided on the UC Berkeley Undergraduate Radio Lab GitHub page and then performed a mean calculation on the data array. A mean was performed on the total dataset to reduce the size and preserve the signal information but in a more manageable format. To store data after calibration, I loaded the data into a '.npz' file with the information on the entry numbers, dates of observation, time stamps for each measurement, b and l coordinates, frequencies, and temperatures. After calculating velocities and distances, I created a separate program to create another '.npz' file to store additional information on the Julian dates, right ascension and declination coordinates, local standards of rest, velocities, and distances.

3.3.2 Median Boxcar Smoothing

Another signal processing method that I used to clean and filter out bad data points from the raw data files is called “Median Boxcar Smoothing”. This is done by running a median across data within a window of points before and after every data point. By taking the median value within a window across all of the data, a new array of “smooth” data points positioned along the shape of the raw signal will be produced. Because the number of elements in our data is approximately 8192, I made the radius of my boxcar 11 points.

In order to remove bad data points, I first subtracted out the smoothed data from the raw data to normalize the signal about the frequency axis. Then, by placing a cutoff range along the power axis above and below four times the standard deviation of the total normalized signal, I was able to find the position of each invalid point that deviates significantly from the standard deviation value. Since both arrays were of the same dimension, the outlier positions should be the same for both datasets. Finally, I replaced the bad data in the raw data with data points taken from the smoothed data.

3.3.3 Polynomial Curve Fitting

While calibrating the spectral lines, we also must remove the baseline from the final product in order to isolate the H1-emission signal. To do this, I used a 5th order polynomial curve to fit to the underlining baseline shape of the data. Additionally, this was done Python using the “PolyFit” function in the Numpy module.

4 Data Analysis

4.1 Temperature Calibration

The initial data points taken are from the radio telescope are recorded in units of power versus frequency. To calibrate the data, I converted the spectral lines into units of temperature with a script written in Python. Before doing this, my partners and I programmed the telescope to record four separate measurements per pointing with equal integration times. As I previously mentioned, the four measurements include a sample with the local oscillator’s frequency set to 1272.4MHz at $10dB_m$, another with the L.O. frequency shifted 4MHz below at 1268.4MHz, $10dB_m$, and two measurements using a noise diode to add $T_{noise} = 100K$ to the total system temperature T_{sys} of each signal. To refer to each of these signals respectfully, the subscript “ONLINE” was set for the higher frequency measurement and “OFFLINE” for the lower one along with “CALON” or “CALOFF” for noise on or off. The CALON measurements, as expected, increased the gain of the signals. Additionally, after examining all four raw data measurements, it was noticed that the gain of the lower 1268.4MHz signal was slightly larger than the gain of the 1272.4MHz signal (see Figure 7 on the following page).

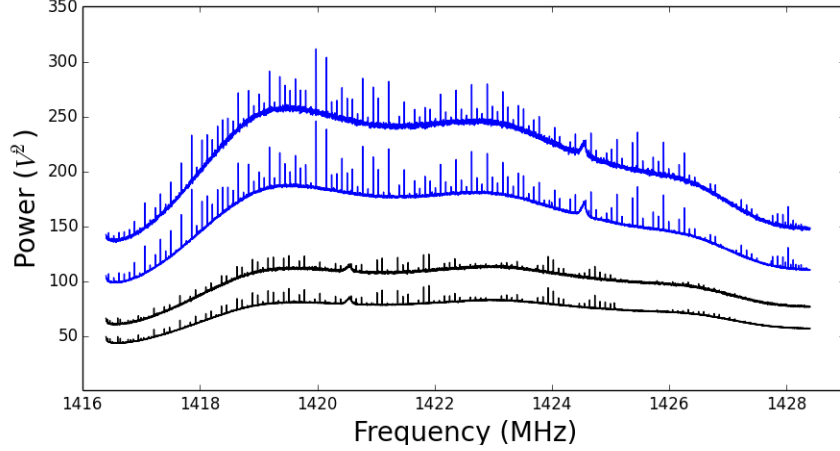


Figure 7: This graph is of the four raw data measurements taken from May 4, 2014 from 19:44:09.18 - 19:48:02.82 UT at $b = -10^\circ$, $l = 75^\circ$. ONLINE measurements at 1272.4MHz are shown in black while OFFLINE measurements at 1268.4MHz are shown in blue. CALON for each ONLINE and OFFLINE measurement is the data shown in the same color above while CALOFF is shown below.

4.1.1 Calibration Process

The first step to calibrate for the spectral lines is to find the system temperature of each of the ONLINE and OFFLINE signals. As shown in Equation 5, by multiplying the noise temperature with the proportionality between the total sum of the CALON measurements and the sum of the difference between the CALON and CALOFF measurements, the system temperature of each measurement can be approximated.

$$T_{sys} = \left[\frac{\sum P_j^{CALON}}{\sum (P_j^{CALON} - P_j^{CALOFF})} \right] \times T_{noise} \quad (5)$$

After, the next step in calibration is to isolate the shape of the ONLINE and OFFLINE, CALOFF signal. To do this, one of the measurements have to be scaled down to the same level as the other. This can be done by multiplying the OFFLINE dataset by the gain factor shown in Equation 6. Figure 8 on the next page shows the result of applying this equation to Figure 7.

$$G = \frac{\sum P_j^{ONLINE, CALOFF}}{\sum P_j^{OFFLINE, CALOFF}} \quad (6)$$

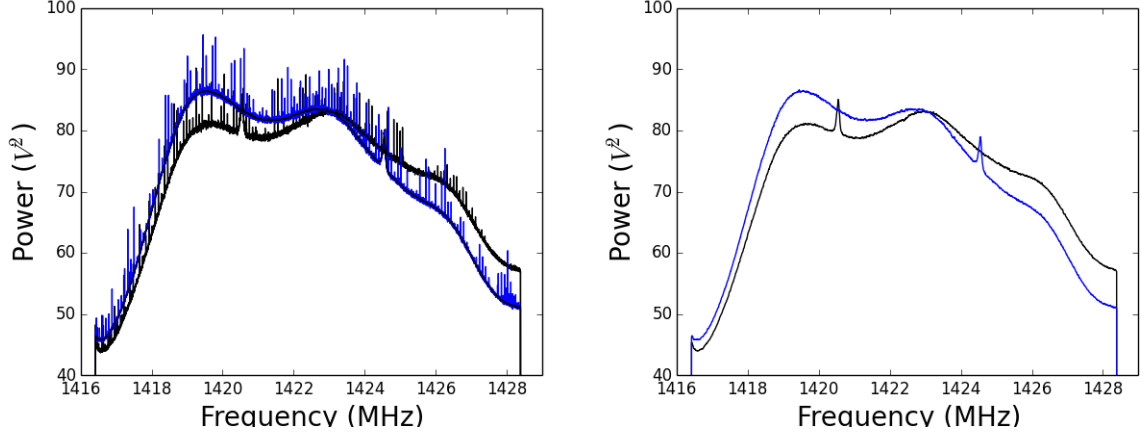


Figure 8: This graph is of the gain conversion and cleaning of data measurements taken from May 4, 2014 from 19:44:09.18 - 19:48:02.82 UT at $b = -10^\circ$, $l = 75^\circ$. ONLINE measurements at 1272.4MHz are shown in black while OFFLINE measurements at 1268.4MHz are shown in blue. Signal cleaning using the method of median boxcar smoothing is shown on the graph on the right. As seen on the figures, the gain conversion is not always perfect. However, this is offset is corrected isolating “shape” of the H1-signal and scaling it with the system temperature.

After setting the ONLINE and OFFLINE, CALOFF measurements to the same level, the next step is to extract the underlining signal “shape”. First, the OFFLINE measurements have to also be shifted back 4MHz to the left so that they aligns with the ONLINE measurements. Next, the section of the data including the H1-emission must be isolated on each set along with the corresponding sections that do not have any H1-emission present. Following Equation 7, by using the “shape” of the H1-signal and scaling it with the system temperature, the data in terms of H1 temperature can finally be calculated.

$$T_{ant,HI}^{ONLINE}(\nu) = \left[\frac{P_{+HI}^{ONLINE}}{P_{none}^{OFFLINE}} \right] \times T_{sys}^{ONLINE}, \quad T_{ant,HI}^{OFFLINE}(\nu) = \left[\frac{P_{+HI}^{OFFLINE}}{P_{none}^{ONLINE}} \right] \times T_{sys}^{OFFLINE} \quad (7)$$

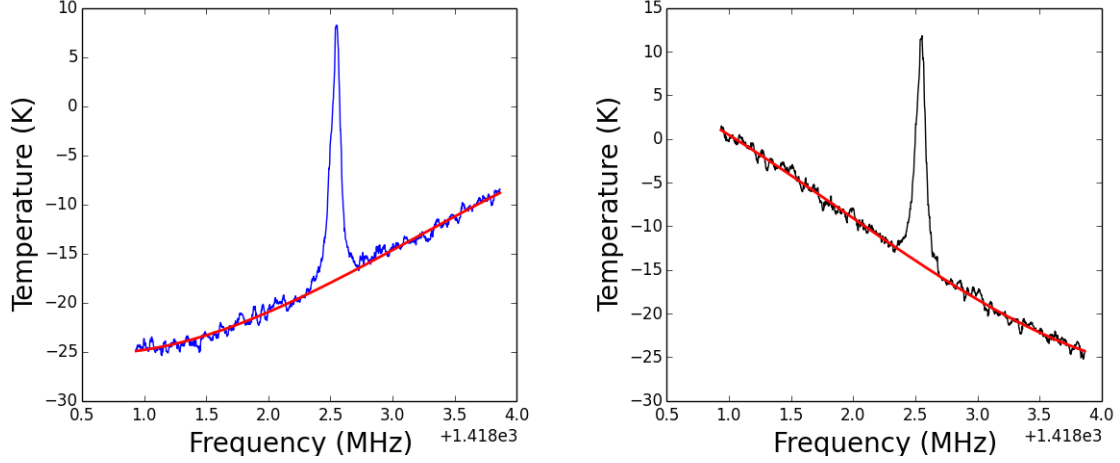


Figure 9: The graph shown above was made from data measurements taken from May 4, 2014 from 19:44:09.18 - 19:48:02.82 UT at $b = -10^\circ$, $l = 75^\circ$. ONLINE measurements at 1272.4MHz are shown in black while OFFLINE measurements at 1268.4MHz are shown in blue. After applying Equation 7 on the previous page to the section of data ranging from 1418.94 - 1421.86MHz, the signals become scaled to units of Kelvin. Because there units in K cannot fall below 0K, a polynomial had to be fit to the data in order to normalize it.

To convert both ONLINE and OFFLINE calibrated temperature spikes into a final spectra, the underlining shape or “base-line” resulting from the artifacts left in the signal must also be removed from the datasets in order to normalize and preserve the H1-emission spectra. This can be done by removing the H1-emission lines from the calibrated temperature datasets and then fitting a polynomial function to the shape of the base-line. Then, by subtracting the polynomial fit from the original ONLINE and OFFLINE temperature calibrated datasets, the H1-emission line on both datasets can be properly normalized. The final step is to then combine and average the two isolated H1-emission datasets (see Equation 8).

$$T_{ant,HI}(\nu) = \frac{1}{2} (T_{ant,HI}^{ONLINE}(\nu) + T_{ant,HI}^{OFFLINE}(\nu)) \quad (8)$$

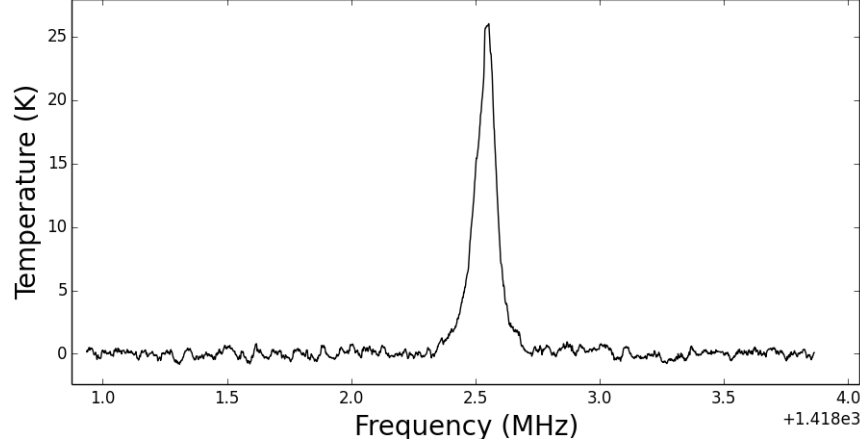


Figure 10: The graph shown above was made from data measurements taken from May 4, 2014 from 19:44:09.18 - 19:48:02.82 UT at $b = -10^\circ$, $l = 75^\circ$ calibrated from power to units of Kelvin. After applying Equation 8 on the previous page to the the difference between the polynomial fit and the two datasets in Figure 9 on the preceding page, we get the final calibrated spectra.

4.1.2 Error Propagation

When taking a long integration and combining spectra, assuming each spectra has statistically the same error ΔT_{sys} , then the error in the sum can be found as shown below in Equation 9.

$$(\text{error in sum})^2 = \left[\sum_0^{N-1} (\Delta T_{sys,n})^2 \right] = N(\Delta T_{sys})^2 \quad (9)$$

$$(\text{error in average})^2 = \frac{(\Delta T_{sys})^2}{N} \quad (10)$$

The average value of a dataset is equal to the sum divided by N . This makes the error associated with averaging squared approximately equal to the error in the sum squared divided by N^2 . Thus the error in averaging is shown by Equation 10. After taking the square root of the equation, we can see how the noise in the data measurements decreases with the square root of N , the number of times that a measurement is repeated. Applying this equation to the general form of Equation 7 on page 10, we get the following expression.

$$\delta T_{sys}(\nu)^2 = \left[\left(\frac{\delta P_j^{ONLINE,CALOFF}}{P_j^{ONLINE,CALOFF}} \right)^2 + \left(\frac{\delta P_j^{OFFLINE,CALOFF}}{P_j^{OFFLINE,CALOFF}} \right)^2 \right] (T_{sys})^2 \quad (11)$$

Since the statistical properties of $P_j^{ONLINE,CALOFF}$ and $P_j^{OFFLINE,CALOFF}$ are the same, we can simplify the expression and combine the contributions of the two spectra (see Equation 12 on the following page). Therefore, the noise in $\delta T_{sys}(\nu)^2$ increases by a multiplicative factor of 2 or the noise in $T_{sys}(\nu)$ by a factor $\sqrt{2}$. Although the ONLINE measurements contain the desired signal information, it is false to assume that spending more time collecting ONLINE than the OFFLINE measurements will optimize the accuracy of the data calibration. The best results are obtained by spending an equal amount of time on each pointing. This is because equal amounts of integration time produces statistically equal amounts of signal noise per pointing. Furthermore, since there is

no way obvious way around inducing some signal noise into our data calibration, it is impossible to generate a completely “noise-free” spectrum.

$$\delta T_{sys}(\nu)^2 = 2 \left(\frac{\delta P_j^{ONLINE,CALOFF}}{P_j^{ONLINE,CALOFF}} \right)^2 (T_{sys})^2 \quad (12)$$

4.2 Cloud Velocities

Since the HI-line frequency is centered about 1420.4MHz, when the line appears at different frequencies, we can conclude that the signals must have been Doppler shifted. Using Equation 1 on page 5, the velocities corresponding to each frequency can be calculated. However, to return accurate Doppler velocity values, we must also correct our measurements to the Local Standard of Rest (LSR). The LSR is a velocity that accounts for the movement of our solar system about the galactic center. In our observation, our LSR correction ranged from -28.9693km/s to -35.7493km/s. Furthermore, to find the velocities corresponding to our line of sight, we must also divide our velocity by the cosine of the inclination angle b (See Equation 13).

$$v_{measured} = (v_{doppler} - LSR) \left(\frac{1}{\cos(b)} \right) \quad (13)$$

Figure 11 shows the result of converting the frequency axis of our dataset into velocities plotted against temperature. For positions within the solar circle that our solar system is approaching (see Figure 5 on page 5), velocities are red shifted while positions outside are blue shifted. On the contrary, for positions within the solar circle that our solar system is receding from, the velocities are blue shifted while the positions outside are red shifted. This red and blue shift is the result of our location direction of motion within the Milky Way.

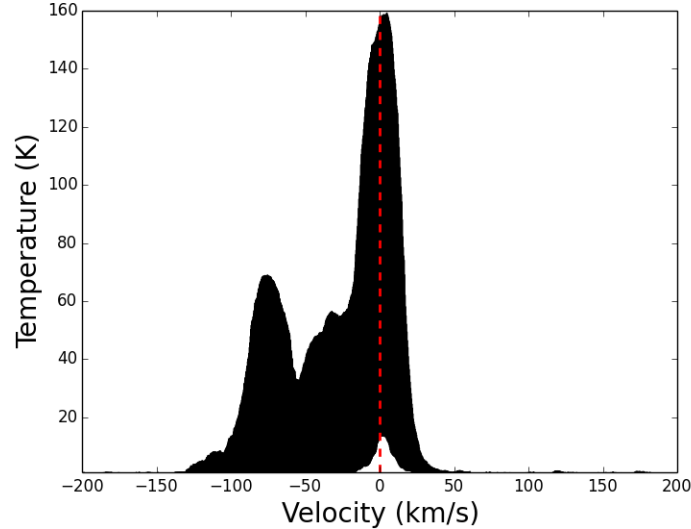


Figure 11: This graph is of all the calibrated data from the May 4, 2014 observation in terms of velocity (km/s) and temperature (K). The data points on the left side of the dashed line are signals that has been red shifted while data points to the right are blue shifted.

4.3 Distance Calculations

To convert our data from velocities into the associated corresponding distances, the angle l , velocity of the Sun about the galactic center $v_0 \approx 220 \text{ km/s}$, and the distance between the two $D_0 \approx 8 \text{ kpc}$ can be utilized. If we form a right triangle between the Earth, the Sun, and positions outside the solar circle, then the angle α can be thought of as the additional angle behind the Sun required to complete 90° (see Equation 14). Additionally, since we know the Sun's velocity, the corrected Doppler velocities, and galactic longitude l , α can be calculated for each dataset.

$$\alpha = \arcsin\left(\frac{v_{\text{measured}}}{v_0} - \sin(l)\right) + l \quad (14)$$

Using angle α , the distance between the Sun and the galactic circle D_0 , and some clever geometry, we can derive a distance equation for our measurements (see Equation 15).

$$D = D_0 [\sin(90^\circ - l) + \sin(l) \tan(90^\circ - \alpha + l)] \quad (15)$$

After converting the velocities into units of distance, my partners and I also solved for the heights above and below the galactic plane by using the distance D and galactic latitude angle b (see Equation 16).

$$H = D \times \tan(b) \quad (16)$$

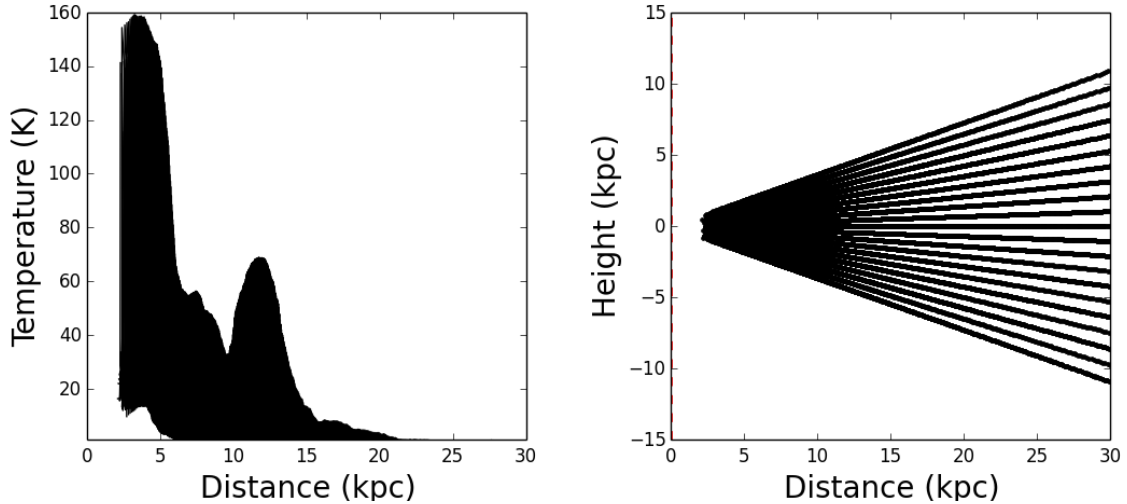


Figure 12: This graph is of all the calibrated data from the May 4, 2014 observation in terms of distance (kpc) and temperature (K) on the left and distance (kpc) and height (kpc) above the galactic plane on the right. As the distances between us and what we are observing increases, we begin to see a drop in the overall temperature. Values ranging from 0 - 3kpc are emissions that has been red shifted. Because of the limited amount of time, my partners and I decided to neglect these values. Additionally, because measurements were taken spaced out in b by 2° , the resolution of our dataset was fairly limited.

4.4 Imaging the Galactic Warp

4.4.1 Generating the Image

Using the final calibrated data, we were able to construct an image. By averaging degenerate datasets, convolving them with a gaussian filter, and zeroing values with intensities below 0K, we were able to map intensity to the second plot shown on Figure 12 on the preceding page. To do this, we first constructed a new image array with x values ranging from 0kpc to 25kpc and y values from -10kpc to 10kpc. Treating each square of information on this picture as a single pixel, the final picture came out to a resolution of roughly 0.1kpc/pix. After, data points were then divided by the resolution to fill every pixel on the array with information with intensities mapped to each pixel.

$$kernel = \exp - \left[\frac{x^2}{2\sigma_x^2} + \frac{y^2}{2\sigma_y^2} \right] \quad (17)$$

The next step was to define weights for the image in order to keep track of the number of data points per pixel and create Gaussian kernel function to convolve or smooth the intensities (See Equation 17). By convolving our image and then dividing it by the weights, we were able to essentially average our data points, bringing them to the corrected levels of intensities. To ensure that the image only convolves meaningful data points, if the pixel of information was not within a range of 3 pixels, the weights were set to 0K.

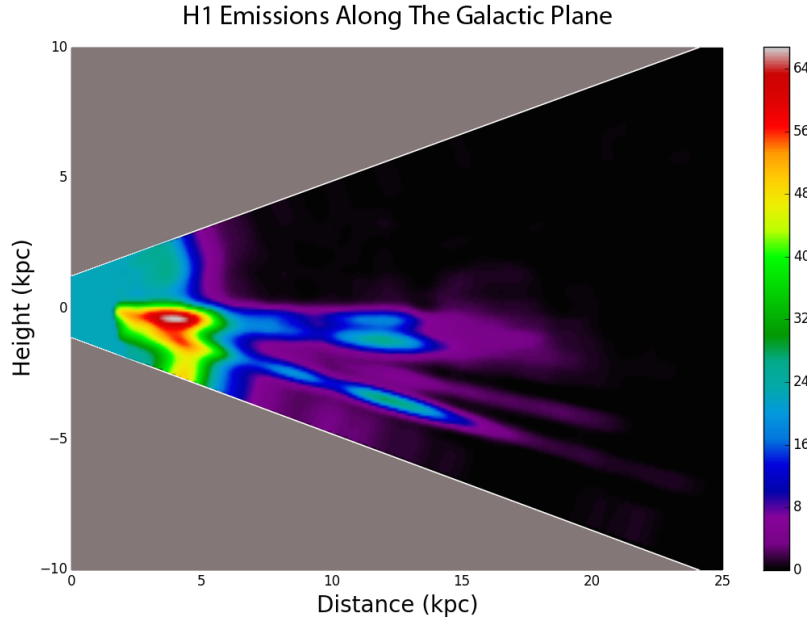


Figure 13: The graph above is a plot all of the data collected from the May 4, 2014, 17:48:37 - 20:41:31 UT observation. Information is plotted in terms distance from our solar system, height above the galactic plane, and temperature intensity in units of Kelvin. on the right. As the distances between us and what we are observing increases, we begin to see a drop in the overall temperature. Additionally, because we took measurements spaced out in b by 2° , the resolution of our dataset shown on the right was limited. For prosthetic purposes, red shifted data points or points between 0-3kpc were filled in with the intensity along the border between red and blue shifts.

4.4.2 Basic Photometric Implications

The optical depth of the H1-emissions along the galactic plane can be taken from the application of Equation 18, where T_{actual} is temperature intensity, τ is the optical thickness, σ_{area} is the cross-sectional area of the H1 emissions, and $N_{emission}$ is the column density or amount of particles per unit of volume.

$$(optical\ depth) = T_{actual}e^{-\tau}, \quad \tau = \sigma_{area}N_{emission} \quad (18)$$

Small values of optical thickness correspond to more information within the visible line of sight. In the case when the optical thickness $\tau \ll 1$, the optical depth reduces to the form $T_{actual}\sigma_{area}N_{emission}$. So, for optically thin regions, the optical depth is larger. When viewing Figure 13 on the previous page, the regions through which H1 temperatures are plotted are optically thin while the lower temperature regions are optically thicker or experience more extinction as a result of dense H1 gas. The region around a distance of 10kpc looks like it may be extinction resulting from the spiral arms of our galaxy. Furthermore, because of the uneven distribution of H1 gas and uneven distributions above and below the galactic plane, Figure 13 on the preceding page supports the concept of galactic warping in the Milky Way. However, additional data collections may be required before a solid conclusion can be derived.

5 Conclusion

5.1 Troubleshooting

Several problems were encountered while trying to carry out this lab. The main issue was with data collection and resulted from an incorrect application of coordinate transformations. This was caught by me when I was calibrating data sets taken from May 1, 2014 to May 3, 2014. If the telescope reaches the boundary of its pointing limits, the H1-line ends up skewed to positions where it should not be. After viewing this happening on a coordinate that was known to be within the pointing limits of the telescope, my partners and I immediately knew something was wrong. After fixing the rotational matrices in our observation code, my partners and I were limited to a 3 hour window on May 4, 2014 for data collection. Other groups performing experiments using the 4.5m radio telescope at the Leuschner Observatory were given more time for their observations based on necessity to complete their report. Since our experiment did not require as much data, the window of observation given to us was already small.

Another issue we noticed was that some of our observations were returning empty values. To ensure this problem did not happen again, we took multiple vertical profiles for our last observation, reducing the resolution of our final image. Nevertheless, with 3 hours of observation time, we were still able to show evidence to support the galactic warp of the Milky Way galaxy.

5.2 Results

With more vertical profiles of the Milky Way at a higher resolution, my partners and I would have been able to make a more detailed image of the galactic warp of the Milky Way galaxy. By plotting multiple profiles and mapping it to the vertical displacement of the H1-emissions, a picture like Figure 14 on the next page could have been constructed. Regardless, I personally am fairly satisfied with the image we constructed in the limited amount of observation time we were given.

Radio telescopes are extremely powerful instruments capable of detecting emissions beyond what is observable from the optical wavelengths. By taking advantage of the 21cm line of H1-gas, regions

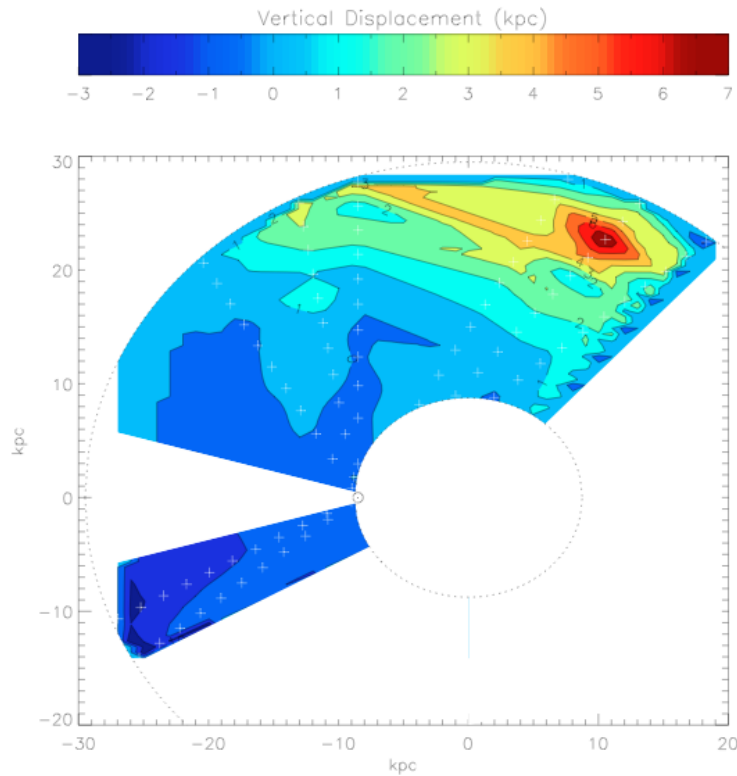


Figure 14: Final galactic warp image created by the undergraduate radio astronomy group Gamma (Andrew, Suneet, James, Nhung) made from data collected using the 4.5m radio telescope at the Leuschner Observatory in 2006. (Image was taken from http://ugastro.berkeley.edu/radio/leuschner/previous.results/gplane_warp.pdf on May 6, 2014.).

of emission can be mapped to temperature intensities, doppler velocities, and distances. Ultimately, this lab has taught how to properly calibrate H1-emission lines, operate the radio telescope at the Leuschner Observatory, derive coordinates for detailed observations, and map plots to positions along regions of space to H1 temperature intensities. Since I want to enter the field of astronomy in the future, the undergraduate radio astronomy course offered at the University of California, Berkeley has been one of the most interesting and important classes for me and my desire to pursue research.

6 Acknowledgements and References

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